

Farm Operations Pipeline Workflow

Google Form -> Google Sheet -> Owner Dashboard

Date: June 16, 2026

1) Objective

- Create a lightweight, reliable system for tracking farm operations in near real-time.
- Make daily logging easy for Armando (mobile-first) while giving the owner a clear dashboard view.
- Support key analysis deliverables: avocado productivity, passionfruit profitability, and labor visibility.

2) End-to-End Workflow

- Step 1: Data capture (Google Forms) - Armando submits harvest, sales, labor, and optional field notes from phone.
- Step 2: Data storage (Google Sheets) - each submission writes to a structured log tab with timestamp.
- Step 3: Data processing - formulas/lookup tables standardize units, enrich rows, and compute derived metrics.
- Step 4: Visualization (Looker Studio) - owner views KPIs, trends, and block/channel performance.
- Step 5: Weekly review - owner + operations check metrics, data quality, and next actions.

3) Core Data Capture Forms (v1)

- A) Avocado Harvest Log: date, block, bins/quantity, unit, crew size, notes.
- B) Passionfruit Sales Log: date, channel, quantity, unit, revenue, price, notes.
- C) Worker Hours Log: date, worker/crew, activity, block/crop area, hours, notes.
- D) Optional Field Note: date, area, category (irrigation/pest/equipment/tree health), note.

4) Google Sheet Structure

- Raw input tabs: harvest_log, passionfruit_sales, labor_log, field_notes.
- Lookup tabs: lookup_blocks, lookup_workers, lookup_channels, lookup_units.
- Processed tabs: standardized tables with computed week/month fields and quality checks.
- Metrics tabs: KPI-ready aggregates for dashboard tiles and charts.

5) Dashboard Design (Owner View)

- Page 1 (Overview): bins this week, passionfruit revenue MTD, labor hours MTD, top sales channel.
- Page 2 (Avocado): bins by block, weekly trend, latest harvest activity.
- Page 3 (Passionfruit): revenue by channel, average price per unit, trendline.
- Page 4 (Labor): hours by activity, hours by area, labor vs harvest proxy.
- Global filters: date range, block, channel, activity.

6) Roles and Responsibilities

- Armando: submit forms daily/weekly and flag operational anomalies in notes.
- Intern (Max): maintain schema, data validation, dashboard logic, and weekly QA checks.
- Owner: review dashboard weekly, align priorities, and approve process adjustments.

7) Implementation Timeline (4 weeks)

- Week 1: discovery and field interview; finalize required fields and constraints.
- Week 2: build forms + sheet structure + lookup tables; test with sample data.
- Week 3: launch pilot; train users; monitor data completeness and friction points.
- Week 4: finalize dashboard; document SOP; present improvement metrics and handoff.

8) Success Metrics

- Data completeness: target $\geq 80\%$ of expected operational events logged.
- Data latency: same-day visibility for harvest/sales activity.
- Owner reporting time: reduce summary prep to < 5 minutes using dashboard.
- Operational transparency: weekly decisions supported by structured data, not memory.

9) Operating SOP (Quick)

- Daily: submit harvest and sales immediately after activity.
- End-of-day or weekly: submit labor hours by activity and area.
- Weekly: review dashboard + data gaps; apply minor schema improvements only after review.

- Monthly: export CSV backup and archive dashboard snapshot for record-keeping.